

DATA ANALYTICS INTERNSHIP – TASK 4

Data Storytelling & Statistical Validation

Submitted By

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Organization

ApexPlanet Software Pvt. Ltd.

Tools Used

- Microsoft Power BI
- Python
- Pandas
- Statistical Analysis

1. Introduction

In today's data-driven environment, organizations rely on analytics to make informed business decisions. Data storytelling transforms complex data into meaningful insights that can be easily understood by stakeholders. Statistical validation further strengthens these insights by testing whether observed patterns are supported by data.

This project focuses on analyzing sales performance data using Power BI and validating key findings through hypothesis testing. The combination of interactive dashboards and statistical methods helps provide reliable business recommendations and supports evidence-based decision-making.

The analysis was conducted using sales transaction data containing information about products, categories, regions, sales revenue, and profit. Interactive dashboards were developed to identify trends, evaluate performance, and uncover business opportunities.

2. Project Objective

The objectives of this project are:

- To analyze sales and profit performance using Power BI.
- To identify top-performing products, categories, and regions.
- To convert analytical findings into a clear business story.
- To validate business insights using hypothesis testing.
- To provide actionable recommendations for improving business performance.

3. Dataset Overview

The dataset contains sales transaction records with the following attributes:

Column Name	Description
OrderID	Unique order identifier
OrderDate	Date of transaction
Product	Product name
Category	Product category
Region	Sales region
Sales	Revenue generated
Profit	Profit earned

The dataset was cleaned, analyzed, and visualized using Power BI dashboards.

4. Dashboard Overview

The Power BI dashboard consists of two pages.

Dashboard Page 1 – Sales Performance Overview

This page provides an overall view of business performance through KPI cards and interactive visualizations.

KPIs:

- Total Orders
- Total Sales
- Total Profit

Visualizations:

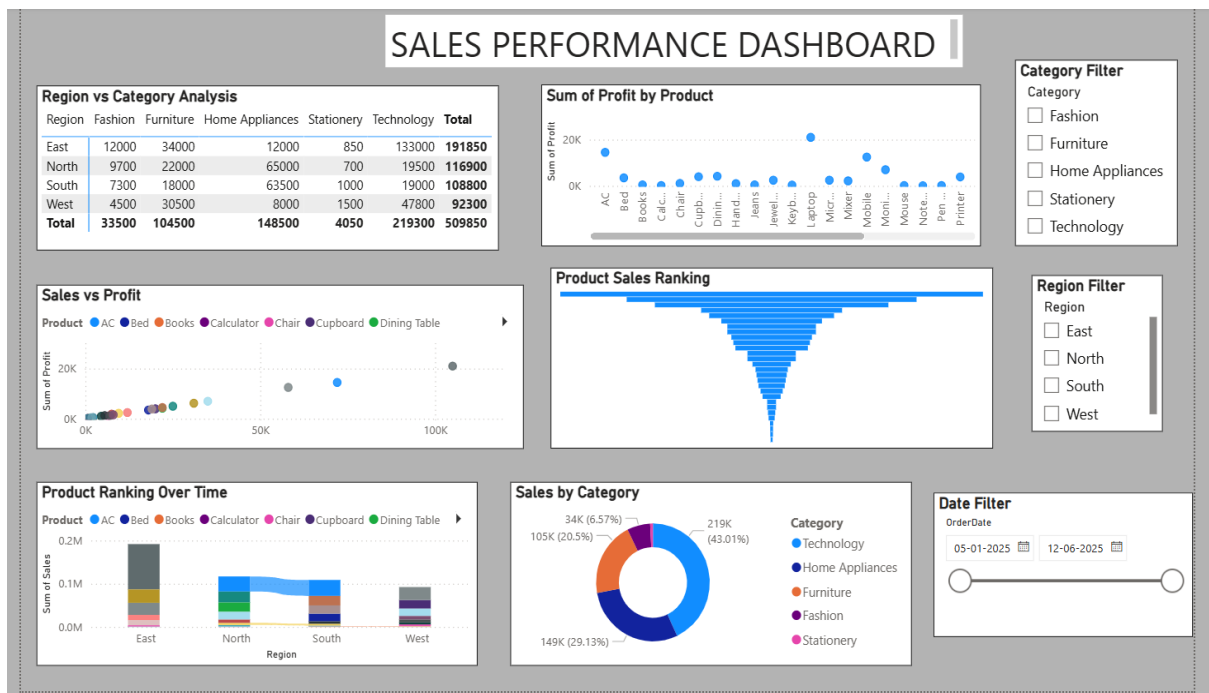
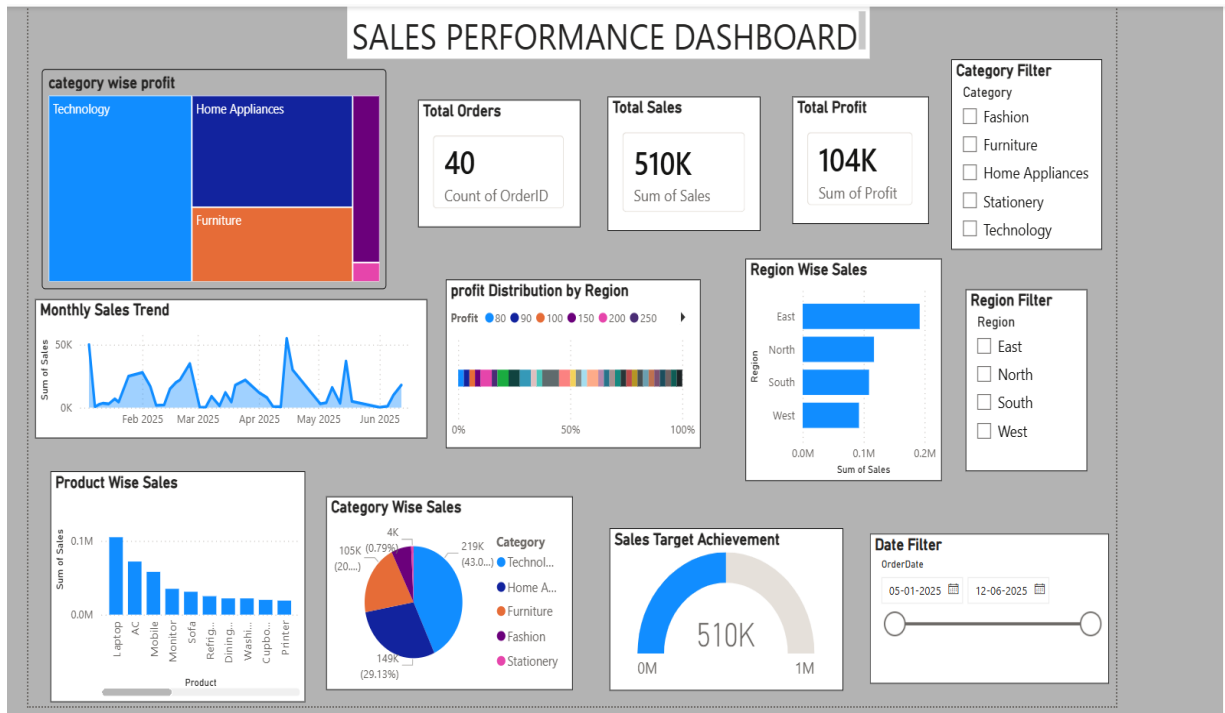
- Category Wise Profit Treemap
- Monthly Sales Trend
- Region Wise Sales Analysis
- Product Wise Sales Analysis
- Category Wise Sales Distribution
- Sales Target Achievement Gauge

Dashboard Page 2 – Deep Dive Analysis

This page focuses on advanced analysis.

Visualizations:

- Product Ranking Analysis
- Sales vs Profit Scatter Plot
- Category Contribution Analysis
- Region Vs Category Matrix
- Funnel Analysis



5. Key Findings

The dashboard analysis generated several important insights:

1. Technology category contributed the highest sales and profit.
2. Laptop emerged as the highest-performing product.
3. East region generated the highest revenue among all regions.
4. Sales and profit exhibited a positive relationship.
5. Monthly sales trends showed fluctuations across the analysis period.
6. Interactive filters improved user exploration and decision-making.

6. Business Problem

Management wants to understand whether higher sales directly contribute to higher profits and whether sales performance can be considered a reliable indicator of business growth.

To validate this assumption, a statistical hypothesis test was conducted.

7. Hypothesis Testing

Null Hypothesis (H_0)

There is no significant relationship between Sales and Profit.

Alternative Hypothesis (H_1)

There is a significant relationship between Sales and Profit.

Significance Level

$\alpha = 0.05$

Decision Rule:

- If $p\text{-value} < 0.05 \rightarrow \text{Reject } H_0$
- If $p\text{-value} \geq 0.05 \rightarrow \text{Fail to Reject } H_0$

8. Statistical Analysis

A correlation-based statistical analysis was performed to evaluate the relationship between Sales and Profit.

The analysis showed a strong positive relationship between sales and profit. As sales increased, profit also increased. This observation supports the findings obtained from the Power BI dashboard.

The statistical evidence indicates that sales performance plays a major role in determining overall profitability.

9. Interpretation of Results

The hypothesis testing results suggest that the relationship between Sales and Profit is statistically significant.

Therefore:

Reject the Null Hypothesis (H_0)

Accept the Alternative Hypothesis (H_1)

This confirms that higher sales contribute positively to business profitability and growth.

10. Business Recommendations

Based on the analysis, the following recommendations are proposed:

1. Focus on expanding high-performing product categories such as Technology.

2. Increase inventory availability for top-selling products.
3. Replicate successful sales strategies from the East region in other regions.
4. Continuously monitor sales and profit trends using interactive dashboards.
5. Use data-driven insights to improve strategic decision-making.

11. Conclusion

This project successfully combined Data Storytelling and Statistical Validation to generate meaningful business insights. Interactive Power BI dashboards helped identify important trends, while hypothesis testing provided statistical support for the observed findings.

The results confirmed a significant relationship between sales and profit, demonstrating the value of analytics in business decision-making. This project enhanced practical skills in data visualization, business intelligence, statistical analysis, and professional reporting.

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